SensorSynth

From Heuristic Biosensor Design to Predictive, Explainable Design Engine

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A tool that helps a researcher answer one question: *I want to detect some molecule in some organism. Which genetic circuit design should I build, and why?*

Goal

A **biosensor** is a living cell (usually a bacterium or yeast) that has been engineered with genetic circuits to detect a specific molecule in its environment and produce a measurable output - like a glow, a color change, or an electrical signal. Building one requires two key decisions:

1. **Which genetic circuit architecture** should detect and respond to the target?
2. **Which host organism (chassis)** should the circuit be built in?

These two decisions are almost never made together in a principled, quantitative way. Sens formalizes this decision process.

Motivation

A 2025 review in *ACS Synthetic Biology* states directly: **"The chassis-design space has remained underexplored as an engineering variable."** This means that despite decades of biosensor research, scientists still largely pick their host organism by habit (usually E. coli) rather than by principled analysis.

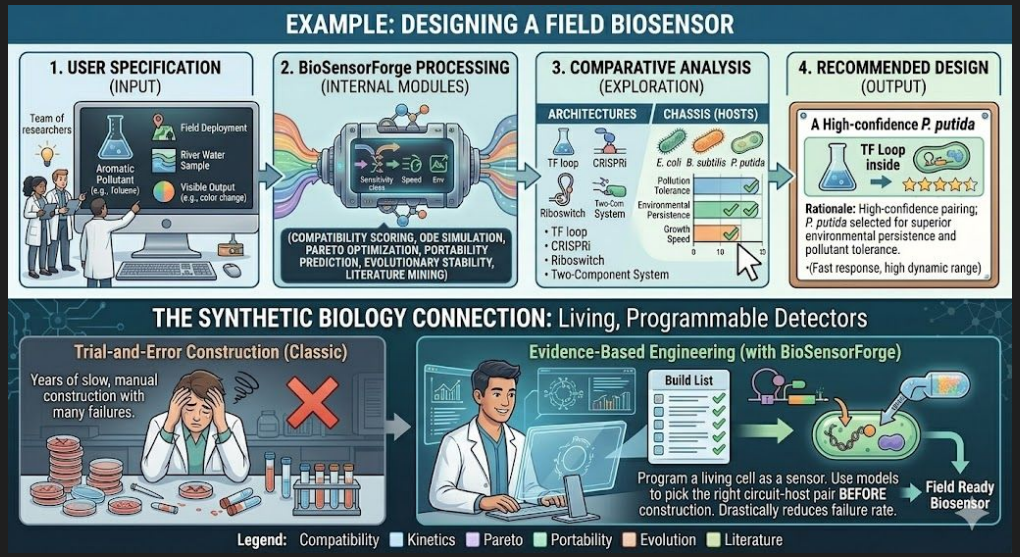
A 2024 paper by Qin et al. in *Quantitative Biology* found that circuit behavior across different organisms follows surprisingly predictable linear relationships — but no tool had yet formalized this into a selection framework.

Sridhar S, explicitly argue that successful biosensor design depends on selecting a chassis based on ecological, metabolic, and genetic constraints, and proposes a framework for systematic chassis choice rather than ad hoc selection.

The Five Open Problems This Project Addresses

Problem	Current State	What it Does
Architecture-chassis mismatch	Discovered after building, wasting months	Predicted before building, computationally
Heuristic performance scores	Arbitrary weights, no mechanistic basis	Hill-function ODE simulation with uncertainty
Single-objective ranking	Pick "best" architecture by one metric	Pareto frontier across 5 simultaneous objectives
Circuit portability prediction	No quantitative framework exists	Transfer coefficient model (Qin 2024 + features)
Circuit evolutionary loss	Rarely considered during design	Stability index predicts half-life in generations

"The same circuit does NOT behave the same across organisms."

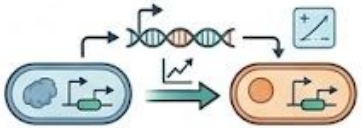


Functionality

Internal Modules: How organte is profilinned iron a design in its on host chassis type

The tool does more than rank options; it explains why a design is good, how fast it should respond, whether it **transfers** to another organism, and how likely it is to fail over generations.

- **Portability model**



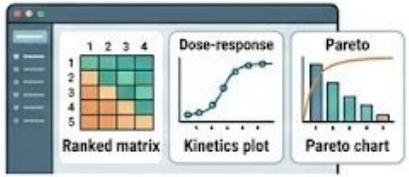
- **Evolutionary stability index**

- **Literature miner**

- **Validation layer**

- **Design need**

- **System produces data**



- **Inspect & Export**

The user can inspect winng one on the dashboard.

OUTPUT 1

Best immediate build

High-confidence architecture in architecture

OUTPUT 2

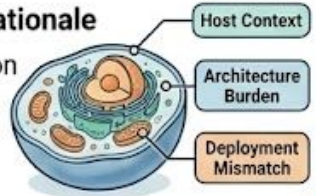
Trade-off alternatives

Backup options options on for user's comparise

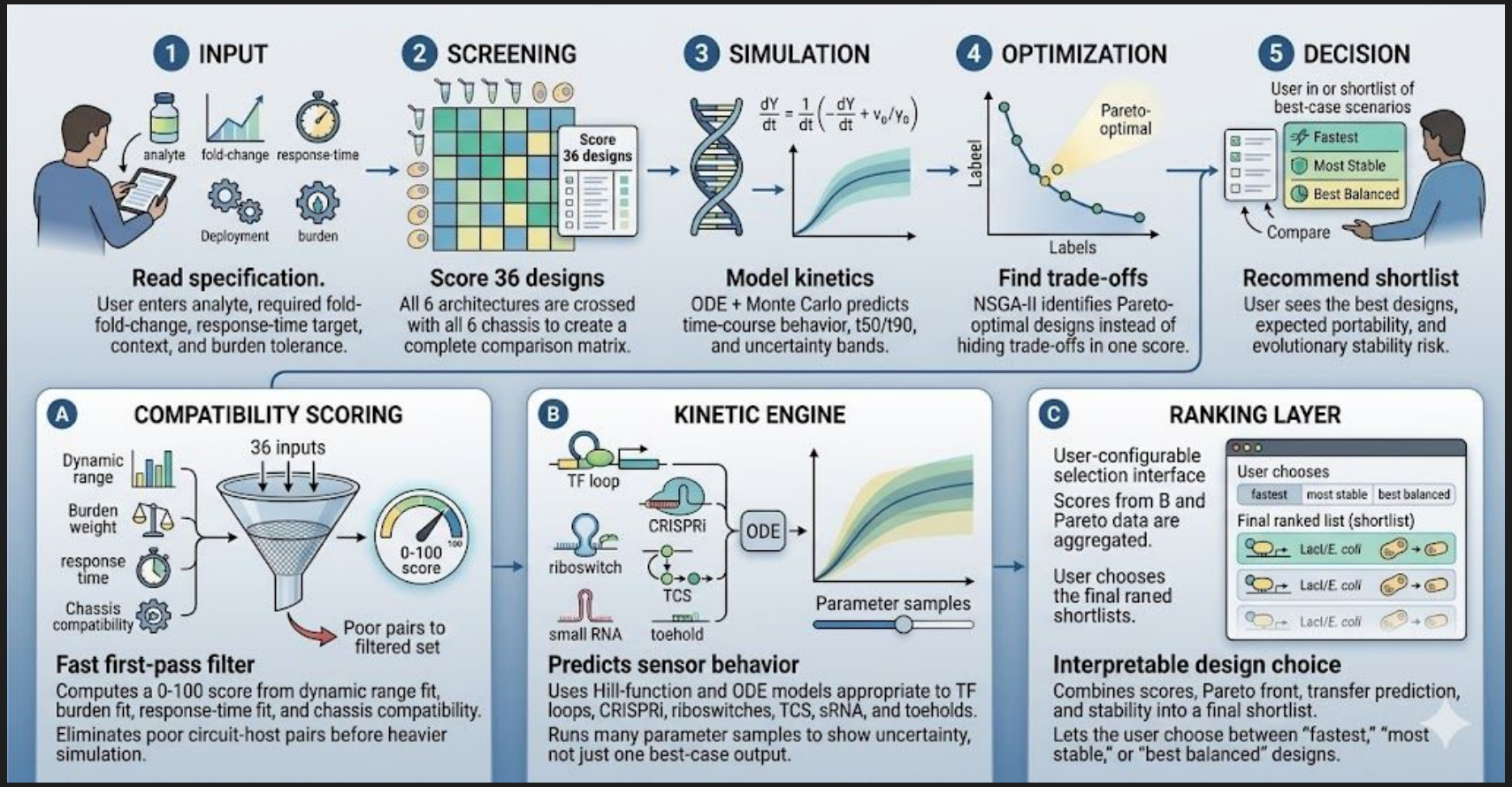
OUTPUT 3

Biological rationale

An explanation is whether provide a full explanation.

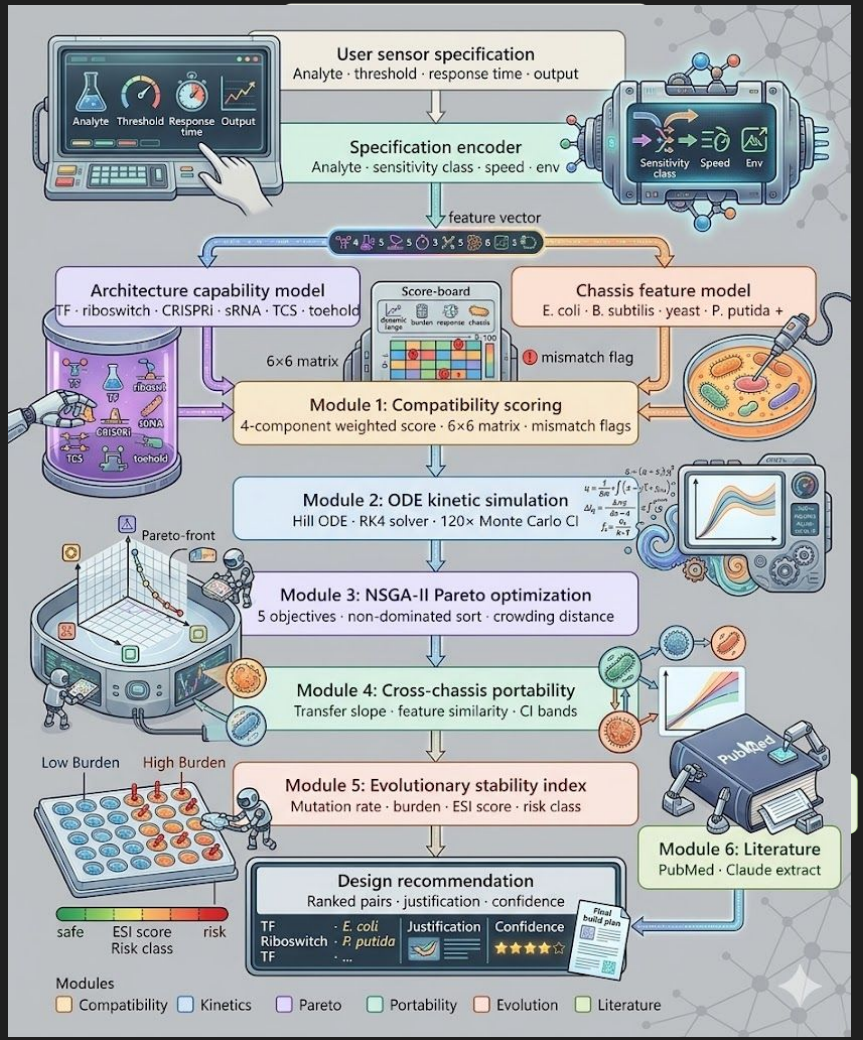


Functionality



Architecture

- **Predictive Portability**: It estimates how well a sensor will perform if moved from one organism to another based on genetic and physical similarities.
- **Long-term Stability**: The system calculates an Evolutionary Stability Index to predict if a circuit will mutate or fail over many generations.
- **Evidence-Based**: Every parameter used in the models is backed by data extracted from primary research literature.
- **Practical Recommendations**: The final output is a ranked shortlist of high-confidence designs with clear biological justifications.



SPECIFICATION

DESIGN SPACE
36
architecture-chassis combinations evaluated
E. coli P. putida Yeast

OBJECTIVES
5
dynamic range, burden, speed, compatibility, stability

MODELS
6
scoring, kinetics, Pareto, portability, evolution, literature mining

PRIMARY USERS
Synthetic biologists

USER OUTPUT
TOP
ranked recommendations with uncertainty and trade-offs

SPECIFICATION REQUIREMENTS (TABLE)

Promoters
Granoters
Promoters

1. ACCEPT SPECIFICATION

analyte/ sensitivity | response time | deployment | burden

Required folding-change

2. COMPARE ARCHITECTURES & HOSTS

Simultaneous Analysis
E. coli
P. putida
Yeast
E. coli
TF loop
CRISPRi
Riboswitch
Acceptable burden
Simultaneous Analysis

3. PREDICT WET-LAB PERFORMANCE

Hill ODE curves
Literature Backed
UNCERTAINTY BANDS
Predictive Monte Carlo Sims

4. RETURN RECOMMENDATIONS
Diverse, optimal designs

Design Plan 1
Kinetic Behavior
Portability
Stability Risk

Plan 2
Kinetic Behavior
Portability
Stability Risk

Trade-offs

Trial-and-Error (Classic):
Years of slow, manual construction with many failures.

Slow, costly construction
Program a living cell as a sensor. → Drastically reduces failure rate.

High-confidence pairing
TF Loop ★★★★★ star-rating justification
P. putida ★★★★★ justification
Final build plan

1. ACCEPT SPECIFICATION | 2. COMPARE ARCHITECTURES & HOSTS | 3. PREDICT WET-LAB PERFORMANCE | 4. RETURN RECOMMENDATIONS

Compatibility | Kinetics | Pareto | Portability | Evolution | Literature

Specification Requirements

SPECIFICATION AREA	REQUIREMENT	CONSTRAINT / TARGET
Input requirements	User must be able to define analyte, dynamic range need, response-time limit, and deployment context.	Required Inputs must map directly to circuit-host scoring.
Prediction quality	System must use mechanistic or evidence-backed models, not arbitrary rankings.	Required Hill/ODE simulation, uncertainty bands, and literature-derived parameters.
Optimization	System must expose trade-offs rather than a single hidden score.	Required Pareto front across 5 objectives.
Biological realism	Must account for chassis compatibility, transferability, and evolutionary loss.	Required Cross-host portability and stability module included.
Usability	Recommendations must be understandable to researchers and fast enough for interactive exploration.	Design constraint Web app workflow, ranked outputs, visual trade-off plots.

Genetic Circuit Architecture

Molecular "wiring" that connects the detection of a molecule to the production of an output.

Architecture 1: TF Promoter-Repressor

Architecture 2: CRISPRi Repression

Architecture 3: Riboswitch

Architecture 4: Small RNA (sRNA) Regulation

Architecture 5: Two-Component System (TCS)

Architecture 6: Toehold Switch

Host Organisms (Chassis)

Chassis 1: Escherichia coli (E. coli)

Chassis 2: Bacillus subtilis

Chassis 3: Saccharomyces cerevisiae (Baker's Yeast)

Chassis 4: Pseudomonas putida

Chassis 5: Corynebacterium glutamicum

Chassis 6: Vibrio natriegens

Detailed Functionality

MODULE 1 — COMPATIBILITY SCORING ENGINE

COMPUTES A 0–100 COMPATIBILITY SCORE FOR ARCHITECTURE × CHASSIS PAIRS GIVEN A SENSOR SPECIFICATION

SCORE = (DynamicRangeFit × 0.35) + (BurdenFit × 0.30) + (ResponseTimeFit × 0.25) + (ChassisCompatFit × 0.10)

WEIGHT JUSTIFICATION: Based on Cardinale & Arkin 2012 (Nat Chem Biol) — survey of biosensor performance metrics frequency in synthetic biology literature.

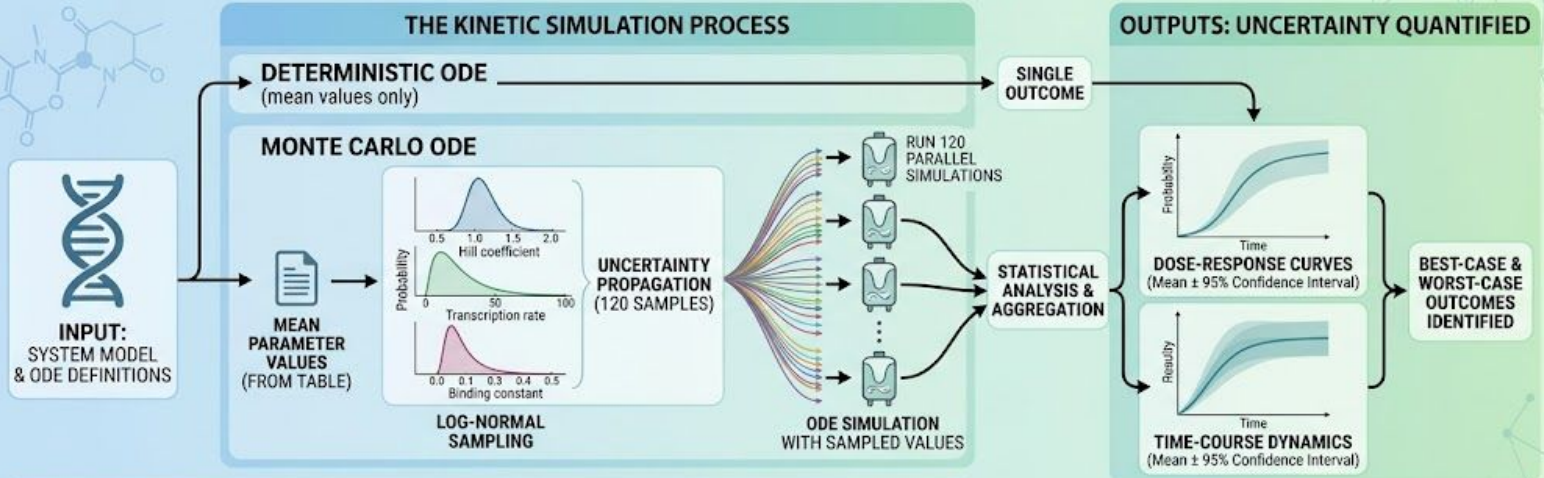
DYNAMIC RANGE FIT	BURDEN FIT	RESPONSE TIME FIT	CHASSIS COMPATIBILITY
Weight: 35%	**Weight:** 30%	**Weight:** 25%	**Weight:** 10%
How it is computed: Compares arch's lit-reported fold-change range to spec's requirement. Penalized if spec needs 100× but arch typically achieves 20×.	**How it is computed:** BurdenFit = chassisBurdenScore − archBurdenCost × (1 − chassisBurdenScore). High-tolerance chassis suffer less penalty for expensive circuits.	**How it is computed:** Arch's typical response time vs. spec's max allowed time. 'Slow' arch (<2h) penalized by 20–65% if spec requires <1h.	**How it is computed:** Domain bonus: +22% for prokaryotic, −38% for eukaryotic (riboswitches). Based on known biological constraints.
Why this weight: Fold-change is the single most-reported metric in biosensor literature	**Why this weight:** Metabolic burden is the most common cause of circuit failure in practice	**Why this weight:** Response speed matters but is more adjustable than dynamic range	**Why this weight:** Fundamental compatibility adds refinement but is not the primary driver

0 10 20 30 40 50 60 70 80 90 100

Detailed Functionality

MODULE 2 — ODE KINETIC SIMULATION WITH MONTE CARLO

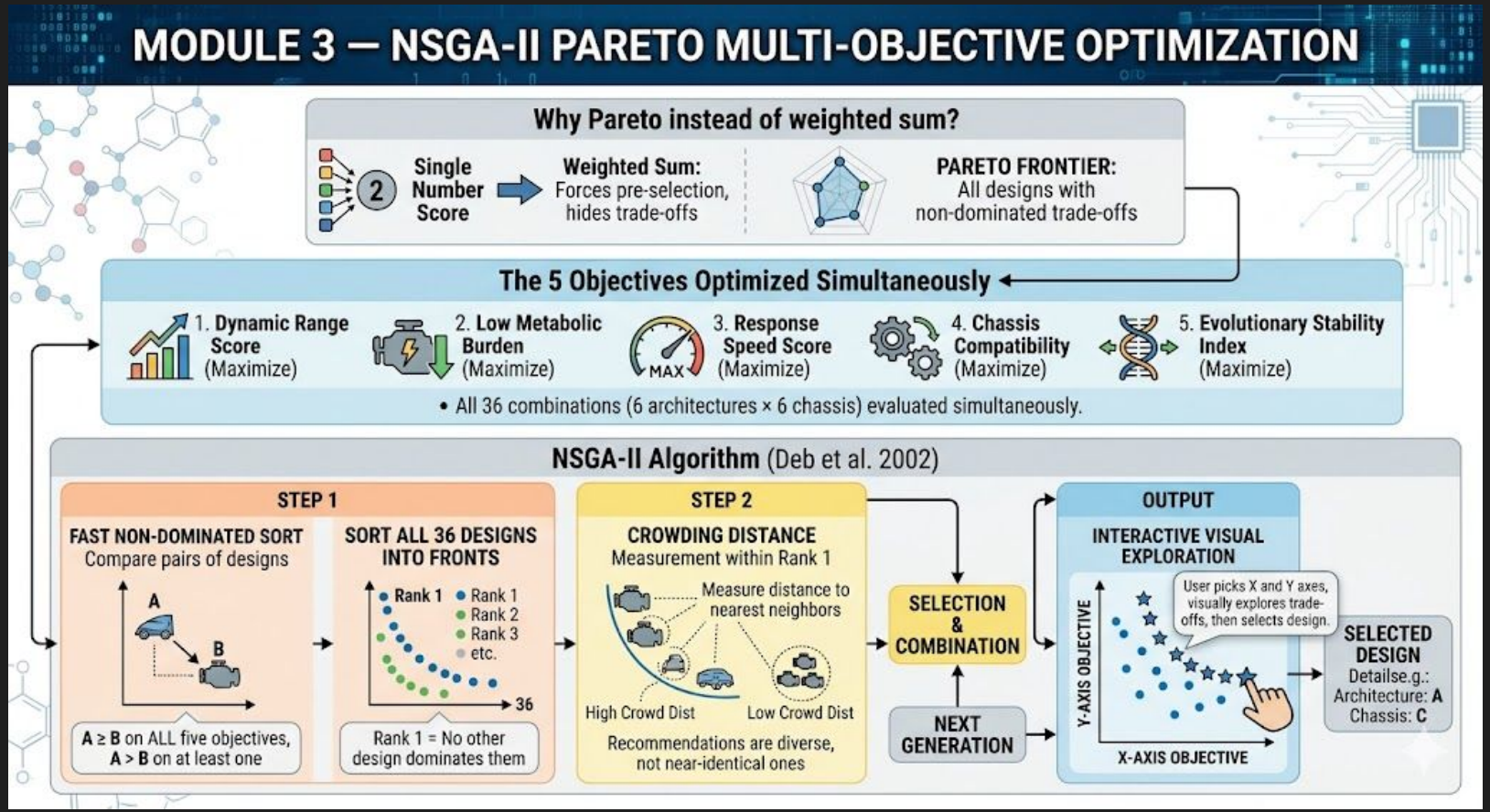
AN EXAMPLE FLOWCHART FOR PROMOTER–OUTPUT DYNAMICS



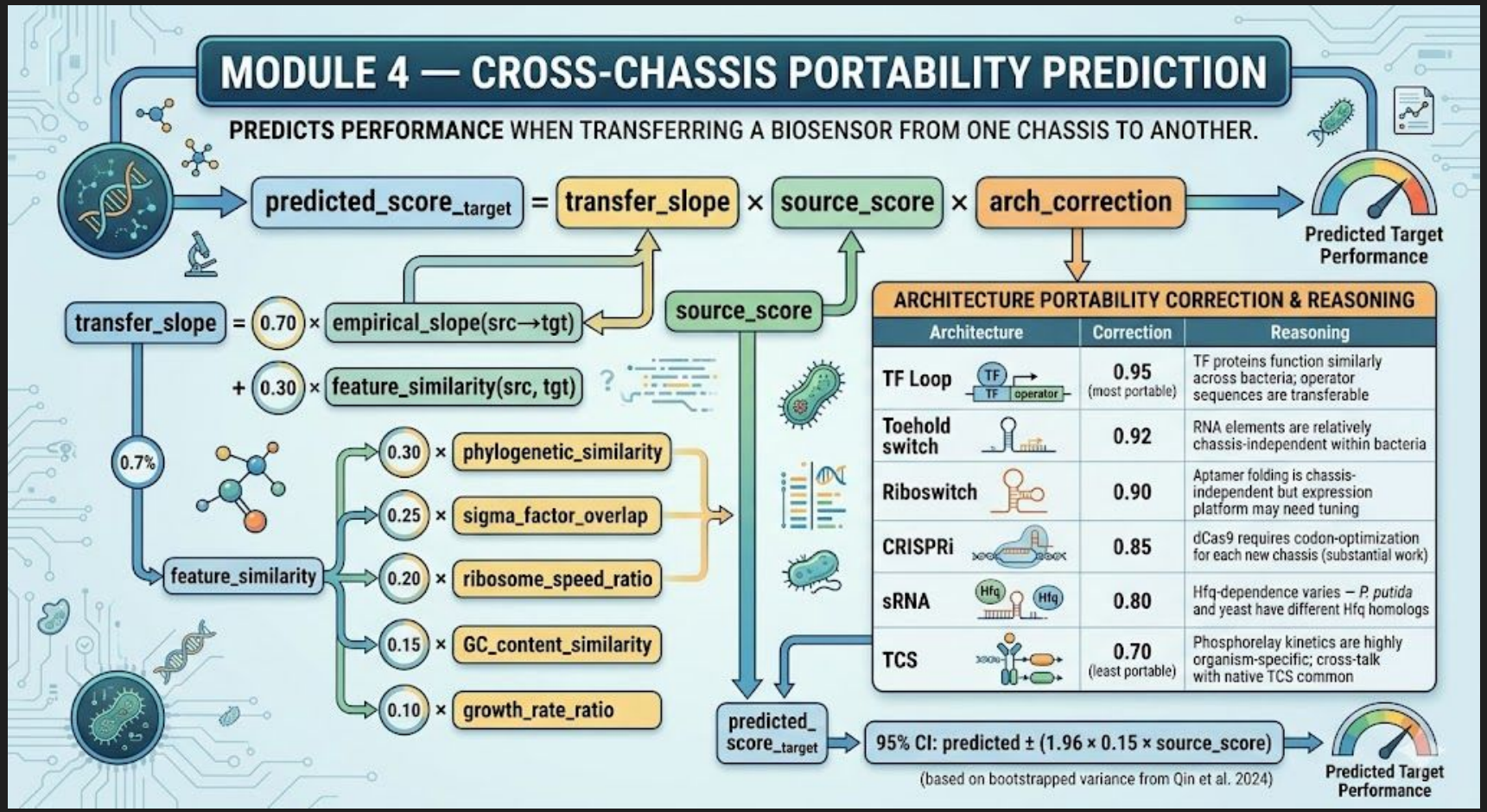
PARAMETER UNCERTAINTY TABLE (*E. COLI* TYPICAL VALUES)

Parameter	Symbol	*E. coli* typical value	CV (uncertainty)	Literature source
Ligand-TF binding constant	K_l	1.0 μM	25%	Barkley 1975
Hill cooperativity	n	2.0	15%	Müller-Hill 1996
Max transcription rate	α_{max}	50 mRNA/h	25%	Proshkin *et al.* 2010
mRNA degradation rate	δ_{mRNA}	13.8 /h	–	Bernstein *et al.* 2002
Protein dilution + deg.	δP	2.1 /h	20%	Calculated from doubling time
CRISPRi K_d	$K_{d\ dCas9}$	0.005 μM	35%	Larson *et al.* 2013
Riboswitch K_d	$K_{d\ apt}$	0.1–10 μM	40%	Wickiser *et al.* 2005

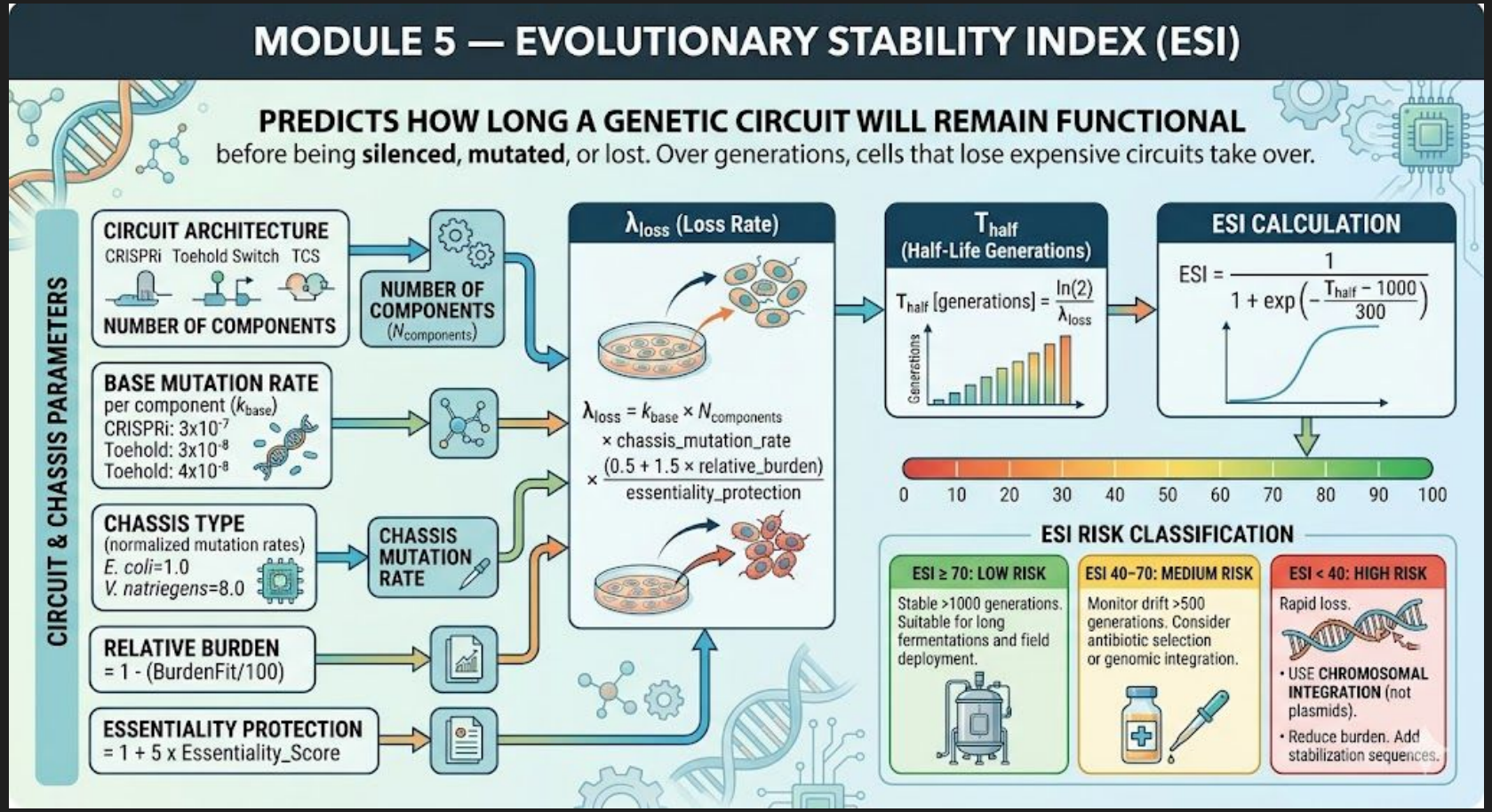
Detailed Functionality



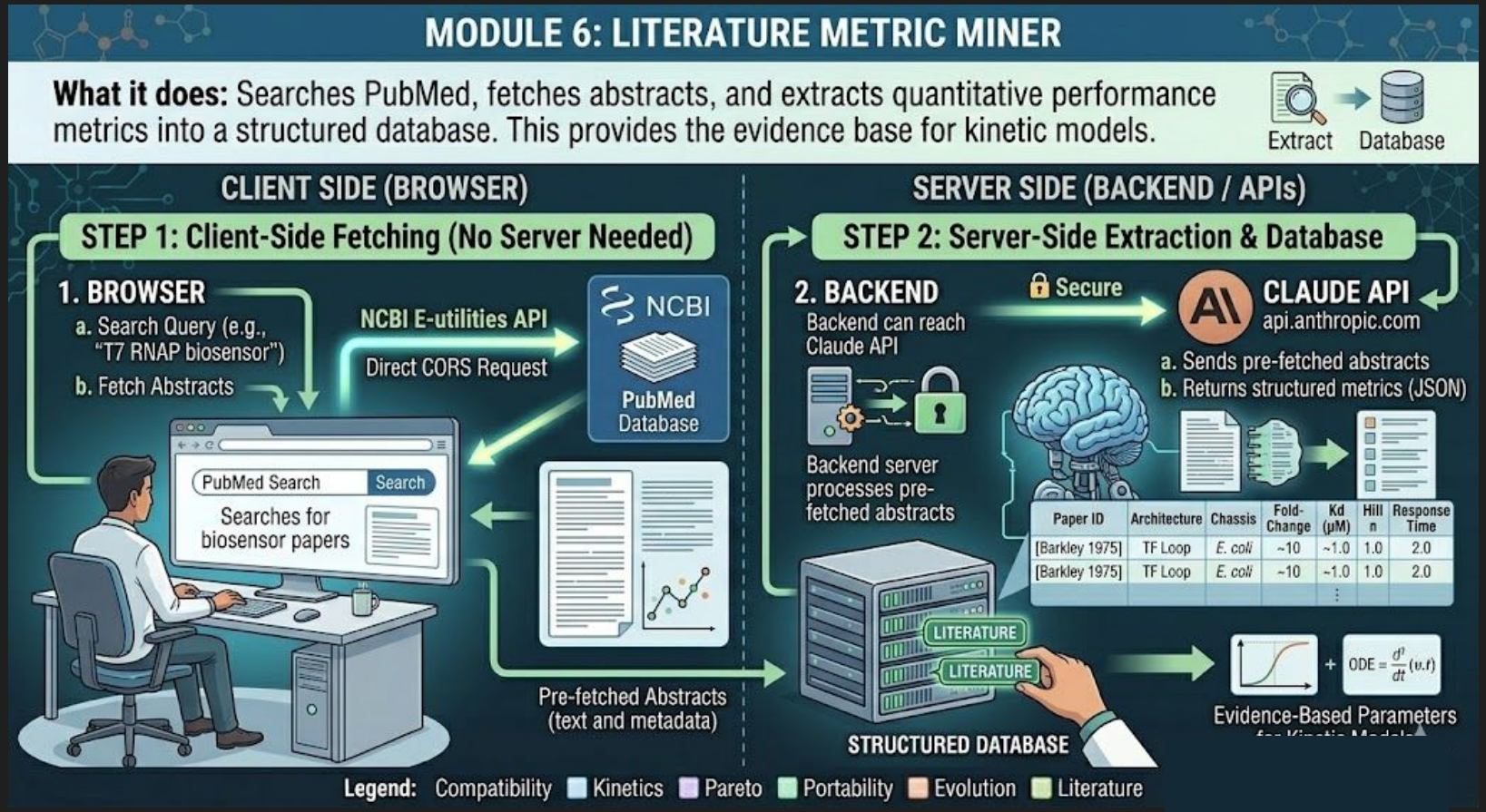
Detailed Functionality



Detailed Functionality



Detailed Functionality



Detailed Example Flow

1 Sensor specification

Define analyte, performance requirements, and host organism. All downstream modules update live.

TARGET ANALYTE

MOLECULE NAME

Arsenic

MOLECULE TYPE

Small molecule (inorganic ion)

DETECTION THRESHOLD

10

UNIT

ppb

OUTPUT TYPE

Fluorescence (GFP/mCherry)

DETECTION ENVIRONMENT

Aqueous solution

PERFORMANCE REQUIREMENTS

REQUIRED FOLD-CHANGE

2 — 20× — 200

MAX RESPONSE TIME (H)

0.5 — 2h — 12

HOST ORGANISM & STRAIN

- Escherichia coli — ×0.33h · high burden
- Bacillus subtilis — ×0.5h · medium burden
- Saccharomyces cerevisiae — ×1.5h · medium burden
- Pseudomonas putida — ×1.2h · high burden
- Corynebacterium glutamicum — ×2h · medium burden
- Vibrio natriegens — ×0.17h · very high burden

STRAIN

ATCC 14048 — Standard wild-type. Increasingly used for rapid biosensor testing.

ATCC 14048 — Standard wild-type. Increasingly used for rapid biosensor testing.

Nat-Bio — Domesticated strain with improved transformation efficiency.

Detailed Example Flow

4 Compatibility score matrix

Architecture × chassis scoring with evolutionary stability. All objectives computed from literature-derived kinetic parameters.

Ranked (6) | Full matrix

#1 Toehold switch

RNA hairpin structure in the 5'-UTR unfolds when a trigger RNA binds, enabling translation of the output gene.

85 / 100 HIGH

DYNAMIC RANGE (35%)	BURDEN FIT (30%)	RESPONSE TIME (25%)	CHASSIS COMPAT (10%)
82	89	90	68

PROMOTERS: Any constitutive promoter

REGULATORS: Trigger RNA-mediated hairpin unfolding (no protein required)

EVOL. STABILITY: 100 / 100

1% ≈ 3,257,271 gen

LOW RISK

EXPLAINABILITY — WHY #1

Toehold switch ranks first for detecting **Arsenic**: high dynamic range capability; metabolic burden is within host tolerance; response kinetics meet the time requirement; evolutionary stability index 100/100 (low loss risk). Suggested next step: retrieve Any constitutive promoter + Trigger RNA-mediated hairpin unfolding (no protein required) from SynBioHub and validate in iBioSim.

#2 Two-component system

Membrane histidine kinase (HK) senses signal and phosphorylates cytoplasmic response regulator (RR) to control gene expression.

81 / 100 HIGH

DYNAMIC RANGE (35%)	BURDEN FIT (30%)	RESPONSE TIME (25%)	CHASSIS COMPAT (10%)
85	85	72	72

PROMOTERS: PompC, PphoA, PphoB

REGULATORS: OmpR, PhoB

EVOL. STABILITY: 100 / 100

1% ≈ 239,016 gen

LOW RISK

#3 CRISPRi repression

78

Detailed Example Flow

4 Compatibility score matrix

Architecture × chassis scoring with evolutionary stability. All objectives computed from literature-derived kinetic parameters.

Ranked (6) | Full matrix

ARCHITECTURE	E. coli	B. subtilis	S. cerevisiae	P. putida	C. glutamicum	V. natriegens
CRISPRi repression	83	64	59	72	59	78
Riboswitch regulation	80	69	61	74	66	77
TF promoter-repressor	76	61	59	68	57	72
Small RNA regulation	80	68	62	73	64	77
Two-component system	85	69	62	76	65	81
Toehold switch	88	77	72	82	73	85

Score = (DynFit×0.35) + (BurdenFit×0.30) + (RespFit×0.25) + (ChassisFit×0.10)

Weights derived from Cardinale & Arkin 2012 biosensor design priority survey (Nat Chem Biol).

Detailed Example Flow

5 ODE kinetic simulation

Hill-function ODEs with Monte Carlo uncertainty quantification. Mean ± 95% CI time-course and dose-response curves. Based on literature-derived kinetic parameters.

Methodology: Instead of heuristic scores, this module simulates actual promoter-output dynamics using Hill-function ODEs (Alon 2007). Monte Carlo sampling (120 parameter sets) propagates kinetic uncertainty (CV ≈ 25–40%) into confidence intervals. t_{50} and t_{90} are computed directly from the kinetic trace.

SIMULATION PARAMETERS

ARCHITECTURE

CRISPRi | Riboswitch | **TF Loop** | sRNA | TCS | Toehold

CHASSIS

Vibrio natriegens

LIGAND CONC. (MM): 1

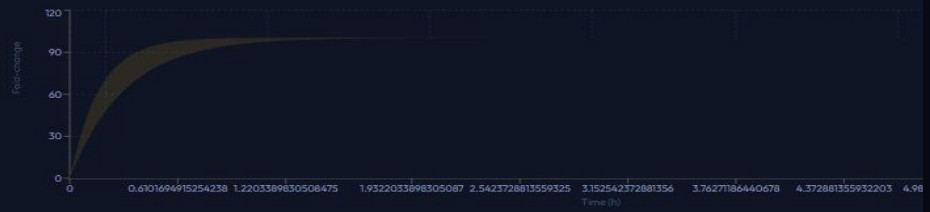
SIM DURATION (H): 6

MC SAMPLES: 120

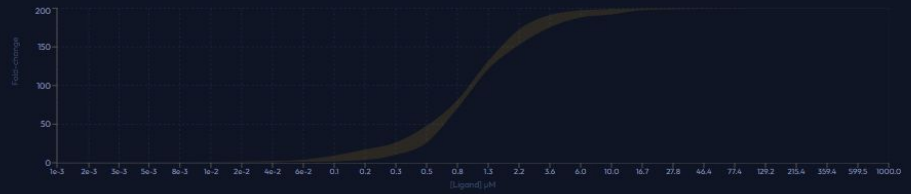
Run ODE + MC (n=120)

100.5×	[100.5, 100.5]×	0.20 h	0.61 h
FOLD-CHANGE (MEAN)	95% CI	T_{50}	T_{90}

TIME-COURSE — FOLD-CHANGE VS TIME (MEAN ± 95% CI, 120 MC SAMPLES)



DOSE-RESPONSE — FOLD-CHANGE VS [LIGAND] MM (LOG SCALE, MEAN ± 95% CI)



Uncertainty reflects literature CV on Kd (±30–40%), Hill n (±20%), and umax (±25%). Shaded region = 5th–95th percentile across 120 MC samples.

Detailed Example Flow

6 Pareto multi-objective frontier

True Pareto-optimal designs across 5 simultaneous objectives. Non-dominated solutions represent the best achievable trade-offs — no weighted sum, no hidden preferences.

Methodology: NSGA-II non-dominated sort (Deb 2002) applied to the architecture–chassis selection problem across 5 objectives: dynamic range, metabolic burden, response speed, chassis compatibility, and evolutionary stability. Stars = Pareto front (rank 1). Crowding distance ensures diverse front coverage.

36	5	31
TOTAL SOLUTIONS	PARETO FRONT SIZE	DOMINATED

X AXIS: Dynamic range | Low burden | Response speed | Chassis compat. | Evol. stability

Y AXIS: Dynamic range | Low burden | Response speed | Chassis compat. | Evol. stability

PARETO FRONT — NON-DOMINATED SOLUTIONS (RANK 1)

DESIGN	DYN RANGE	LOW BURDEN	RESPONSE	COMPAT	EVOL STAB	SCORE	CROWDING
CRISPRi repression Escherichia coli	90	100	62	62	100	83	∞
Riboswitch regulation Escherichia coli	58	100	92	72	100	80	∞
Two-component system Escherichia coli	85	100	72	72	100	85	∞
Small RNA regulation Escherichia coli	60	100	88	70	100	80	1.75
Toehold switch Escherichia coli	82	100	90	68	100	88	1.71

Detailed Example Flow

7 Cross-chassis portability

Predict circuit performance when transferred to a different host. Based on Qin et al. 2024 empirical linear transfer model extended with feature-based similarity.

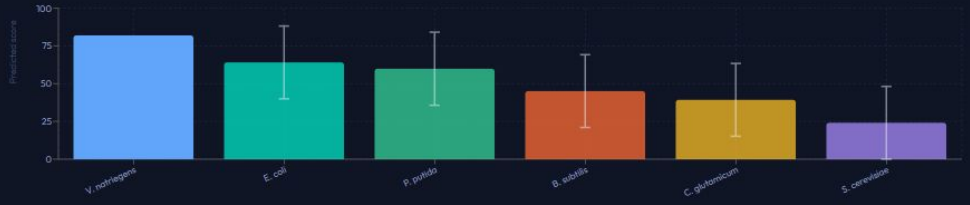
Methodology: Cross-chassis portability prediction is an open problem. No existing tool formalizes this transfer. Our model blends the empirical linear slope matrix (Qin et al. 2024 Quantitative Biology) with a 5-feature similarity vector (phylogeny, sigma factor overlap, ribosome speed, GC content, growth rate). Architecture-specific correction accounts for codon optimization requirements.

TRANSFER PREDICTION SETUP

ARCHITECTURE	SOURCE CHASSIS (CHARACTERIZED IN)	KNOWN PERFORMANCE SCORE IN SOURCE
TF promoter-repressor	Vibrio natriegens	82

Predict portability across all chassis

PREDICTED PERFORMANCE: TF PROMOTER-REPRESSOR TRANSFERRED FROM VIBRIO NATRIEGENS (SCORE=82)



PORTABILITY DETAIL TABLE

TARGET CHASSIS	PREDICTED	95% CI	TRANSFER SLOPE	FEATURE SIM	CONFIDENCE	NOTE
Vibrio natriegens	82.0	[82.0, 82.0]	1.000	—	HIGH	Same chassis — no transfer needed.
Escherichia coli	64.1	[40.0, 88.2]	0.781	0.688	HIGH	Based on Qin et al. 2024 linear transfer model with tf_loop correction.
Pseudomonas putida	59.9	[35.8, 84.1]	0.731	0.605	MEDIUM	Based on Qin et al. 2024 linear transfer model with tf_loop correction.
Bacillus subtilis	45.1	[21.0, 69.2]	0.550	0.484	MEDIUM	Based on Qin et al. 2024 linear transfer model with tf_loop correction.
Corynebacterium glutamicum	39.3	[15.2, 63.4]	0.479	0.398	MEDIUM	Based on Qin et al. 2024 linear transfer model with tf_loop correction.
Saccharomyces cerevisiae	24.1	[0.0, 48.2]	0.293	0.213	LOW	Based on Qin et al. 2024 linear transfer model with tf_loop correction.

6 Literature metric miner

Search PubMed, fetch abstracts directly from your browser, then extract quantitative biosensor metrics. Add an Anthropic API key for AI-powered extraction, or use heuristic mode without one.

How it works: Your browser fetches articles directly from NCBI PubMed (no server proxy needed). Abstracts are then sent to our backend where Claude extracts structured data — fold-change values, binding constants (Kd), Hill coefficients, response times, and chassis used. This builds the evidence base that justifies our kinetic model parameters.

QUICK SEARCH PRESETS

CRISPRi sensors | Riboswitch performance | TF-based sensors | sRNA regulation | Two-component systems | Toehold switches | Non-model chassis | Chassis compatibility | Metabolic burden | Evolutionary stability

PUBMED SEARCH QUERY

riboswitch biosensor sensitivity bacteria response time

MAX PAPERS: 8

ANTHROPIC API KEY (OPTIONAL — ENABLES AI EXTRACTION)

sk-ant-... (leave blank for heuristic extraction)

Without an API key, pattern-based heuristic extraction is used. With a key, Claude identifies metrics more accurately.

Search & extract (8 papers)

PER-PAPER EXTRACTION RESULTS (2 PAPERS)

Modeling a synthetic aptamer-based riboswitch biosensor sensitive to low hexahydro-1,3,5-trinitro-1,3,5-triazine (RDX) concentrations. RIBOSWITCH 15% CONF

Mayo ML, Eberly JO, Crocker FH · *PLoS one* (2020) · PMID 33253235

RNA aptamers are relatively short nucleic acid sequences that bind targets with high affinity, and when combined with a riboswitch that initiates translation of a fluorescent reporter protein, can be used as a biosensor for chemical detection in various types ...

In: E. coli

Pattern-based extraction — add an Anthropic API key for accurate LLM extraction.

Translational Enhancer Based Amplification of Toehold Sensors (TacToe) for Improved Sensitivity and Speed of Viral RNA Detection. 0% CONF

Kale T, Pednekar R, Cazaux SM · *ACS synthetic biology* (2025) · PMID 40439353

Abstract not retrieved.

⚠ No abstract provided

Detailed Example Flow

9 Evolutionary stability & validation

Predict circuit half-life under natural selection. Concordance analysis against 10 published experimental outcomes.

Evolutionary stability | Validation / concordance

Methodology: Evolutionary stability prediction is rarely formalized. We model circuit half-life as a function of metabolic burden, component count, mutation rate, and essentiality. Based on Sleight et al. 2010 (J Biol Eng) burden-driven loss model and Helenek et al. 2024 mid-scale evolution theory (Chem Biol).

CRISPRi repression — LOW RISK
Escherichia coli
STABILITY INDEX 100
t½: 1,540,327 gen | 508308 h

CRISPRi repression — LOW RISK
Bacillus subtilis
STABILITY INDEX 100
t½: 175,836 gen | 87918 h

CRISPRi repression — LOW RISK
Saccharomyces cerevisiae
STABILITY INDEX 100
t½: 482,861 gen | 724292 h

CRISPRi repression — LOW RISK
Pseudomonas putida
STABILITY INDEX 100
t½: 148,108 gen | 177730 h

CRISPRi repression — LOW RISK
Corynebacterium glutamicum
STABILITY INDEX 100
t½: 222,591 gen | 445181 h

CRISPRi repression — LOW RISK
Vibrio natriegens
STABILITY INDEX 100
t½: 125,027 gen | 21255 h

ALL DESIGNS — STABILITY RANKING

ARCHITECTURE	CHASSIS	ESI	T½ (GEN)	T½ (H)	RISK	BURDEN	MUTATION TARGETS
CRISPRi repression	*Escherichia coli*	100	1,540,327	508308	LOW	0%	3
CRISPRi repression	*Bacillus subtilis*	100	175,836	87918	LOW	64%	3
CRISPRi repression	*Saccharomyces cerevisiae*	100	482,861	724292	LOW	73%	3
CRISPRi repression	*Pseudomonas putida*	100	148,108	177730	LOW	36%	3
CRISPRi repression	*Corynebacterium glutamicum*	100	222,591	445181	LOW	82%	3
CRISPRi repression	*Vibrio natriegens*	100	125,027	21255	LOW	18%	3
Riboswitch regulation	*Escherichia coli*	100	27,725,886	9149542	LOW	0%	1
Riboswitch regulation	*Bacillus subtilis*	100	4,318,674	2159337	LOW	38%	1

Successes

Scientific success

Literature-grounded models

Each module was tied to biosensor, gene circuit, and synthetic biology literature, so the tool was not just conceptual software.

Engineering success

Working full-stack design

The backend, models, API endpoints, and frontend pages formed a complete product rather than a disconnected theoretical proposal.

Practical success

Real decision support

Outputs such as ranked matrices, kinetics plots, Pareto fronts, and stability scores are directly useful during design planning.

OVERALL CONCLUSION OF SUCCESS

- **It transformed an open-ended design problem into a formal workflow:** researchers can now move from "What should I build?" to a justified shortlist.
- **It balanced novelty and feasibility:** the portability and stability modules add innovation, while the compatibility and kinetics modules keep the system grounded in known biology.
- **It created a strong platform for future work:** mammalian chassis, flux balance analysis, and live literature updates can all build naturally on the current architecture.

End-to-End Pipeline

From literature to design recommendation — fully automated

Data-Driven Decisions

Every suggestion backed by extracted evidence, not intuition

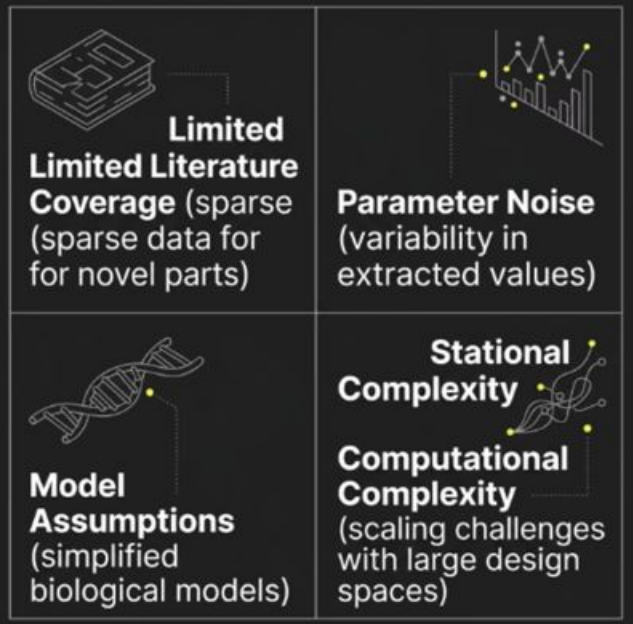
Explainability

Full traceability of why each design was ranked as it was

Real Implementation

Validated against known working biological systems

Limitations & Failure Modes



1. Literature Gaps

- **Issue:** Limited data for novel parts
- **Observed:** Missing parameters → fallback assumptions
- **Fix:** Integrate full-text extraction + experimental datasets

2. Model Simplifications

- **Issue:** Assumes ideal cellular context
- **Observed:** Overestimated performance in complex environments
- **Fix:** Incorporate context-aware / stochastic models

3. Parameter Noise

- **Issue:** Variability in extracted metrics
- **Observed:** High variance in Monte Carlo outputs
- **Fix:** Confidence-weighted extraction + filtering

4. Compute Scaling

- **Issue:** Large design space
- **Observed:** NSGA-II runtime increased significantly
- **Fix:** Use surrogate models / pruning strategies

Future Work

1. **Wet-Lab Validation**
   Close the loop with experimental testing of top-ranked designs
2. **Improved Biological Models**
   Incorporate context-dependent expression and host-specific effects
3. **RL Optimization**
   Use reinforcement learning to iteratively refine design recommendations
4. **Expanded Database**
   Scale literature coverage and integrate community-submitted data

Additional:

The PubMed pipeline was completely rearchitected. But, it still cannot retrieve all the information about chassis and the genetic circuit.

The old design had the server fetch from NCBI which fails in restricted server environments. The new design has your browser fetch directly from NCBI (no restrictions, CORS is allowed) and only sends the pre-fetched abstracts to the backend.

DISCUSSION